



US00PP34737P2

(12) **United States Plant Patent**
Blaker et al.

(10) **Patent No.:** **US PP34,737 P2**
(45) **Date of Patent:** **Nov. 15, 2022**

(54) **STRAWBERRY PLANT NAMED**
‘CONSTITUTION’

(50) Latin Name: *Fragaria ananassa*
Varietal Denomination: **Constitution’**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/377,177**

(22) Filed: **Jul. 15, 2021**

(51) **Int. Cl.**
A01H 5/08 (2018.01)
A01H 6/74 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./209**
CPC **A01H 6/7409** (2018.05)

(58) **Field of Classification Search**
USPC Plt./209
CPC A01H 6/7409; A01H 5/08
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP20,394 P3 10/2009 Ackerman et al.
PP30,564 P3 6/2019 Whitaker

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of
strawberry plant named ‘Constitution’. This new strawberry
plant named ‘Constitution’ is primarily adapted to the grow-
ing conditions of West Central Florida, and is primarily
characterized by its achenes typically set even with or just
below the surface of the fruit, vigorous plant habit, very high
marketable yield, early time of first flower and fruit, and
large berry size.

4 Drawing Sheets

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Latin name of the genus and species of the plant claimed:
Fragaria ananassa.

Variety denomination: ‘CONSTITUTION’.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct straw-
berry variety named ‘CONSTITUTION’. This new variety
is a result of a controlled cross made in 2012 in an ongoing
breeding program between the unpatented strawberry vari-
ety designated ‘Red Merlin’ as the seed (female) parent, and
the unreleased, unpatented strawberry breeding selection
designated ‘BG-5.321’ as the pollen (male) parent. The
variety is botanically known as *Fragaria ananassa*.

The seedling resulting from the aforementioned cross was
selected from a controlled breeding plot in Hillsborough
County, Fla. in the fall/winter of 2013-2014. After its
selection, the new variety was asexually propagated by
stolons in both Siskiyou County, Calif. and San Joaquin
County, Calif. The new variety was extensively tested over
the next several years in fruiting fields in Hillsborough
County, Fla. This propagation has demonstrated that the
combination of traits disclosed herein as characterizing the
new variety are fixed and remain true-to-type through suc-
cessive generations of asexual reproduction.

BRIEF SUMMARY OF THE INVENTION

‘CONSTITUTION’ is primarily adapted to the climate
and growing conditions of West Central Florida. The sub-
tropical climate of West Central Florida provides the day

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length and moderate temperatures needed to produce an
early yielding, vigorous plant and maintain fruit quality
during the fall and winter production months.

The following traits have been repeatedly observed and
are determined to be unique characteristics of ‘CONSTITI-
TUTION’, which in combination distinguish this strawberry
plant as a new and distinct variety:

1. Achenes typically set even with or just below the
surface of the fruit;
2. Vigorous plant habit;
3. Very high marketable yield;
4. Early time of first flower and fruit; and
5. Large berry size.

‘Florida Brilliance’ (U.S. Plant Pat. No. 30,564) is cur-
rently the dominant strawberry variety in Hillsborough
County, Fla. The fruits of ‘CONSTITUTION’ are similar in
color, flavor, and firmness to ‘Florida Brilliance’, but the
fruits of ‘CONSTITUTION’ are more elongated and the
plant is more vigorous. The achenes of ‘Florida Brilliance’
are slightly more sunken than those of ‘CONSTITUTION’
and its plant architecture is more open. In side-by-side
comparisons from the 2020-2021 season (Nov. 19, 2020 to
Feb. 19, 2021) ‘CONSTITUTION’ compares with ‘Florida
Brilliance’ (U.S. Plant Pat. No. 30,564) in the following
combination of characteristics as described in Table 1.

TABLE 1

Characteristic	'SB_12_101-108'	'Florida Brilliance' (U.S. Plant Pat. No. 30,564)
November marketable yield (gm/plt)	54	20
Season marketable yield (gm/plt)	273	269
November average berry size (gm)	18.2	12.7
Season average berry size (gm)	21.1	24.0
Average runners/plant	1.9	1.2

For identification, a series of molecular markers have been determined for this new variety.

'CONSTITUTION' differs from its parents, 'Red Merlin' and 'BG-5.321' by the following combination of characteristics as described in Tables 2 and 3.

TABLE 2

Characteristic	'SB_12_101-108'	'Red Merlin'
Fruit: size	Large	Medium
Fruit: marketable yield	Very high	Medium
Fruit: flavor	Fair	Fair to good
Fruit: seed position	Slightly below to even with surface	Even with surface

TABLE 3

Characteristic	'S6_12_101-108'	'BG-5.321'
Fruit: size	Large	Very large
Fruit: marketable yield	Very high	High
Fruit: early yield	Very high	High
Fruit: continuous yielding	Gaps mid-season	Continuous

BRIEF DESCRIPTIONS OF THE PHOTOGRAPHS

The accompanying color photographs illustrate the overall appearance of typical specimens of the new strawberry variety 'CONSTITUTION' at various stages of development, as true as it is reasonably possible with color reproductions of this type. Color in the photographs may differ slightly from the color value cited in the botanical descriptions which accurately describe the color of 'CONSTITUTION'. The depicted plant and plant parts of the new strawberry variety 'CONSTITUTION' are approximately four months old. The photographs were taken in Hillsborough County, Fla.

FIG. 1 shows typical fruiting field characteristics of 'CONSTITUTION', taken in the month of February 2021;

FIG. 2 shows a close-up view of a typical plant of 'CONSTITUTION', taken in the month of February 2021;

FIG. 3 shows typical mature and immature field fruit of 'CONSTITUTION', taken in the month of February 2021; and

FIG. 4 shows typical internal and external mature fruit characteristics of 'CONSTITUTION', taken in the month of February 2021.

DETAILED BOTANICAL DESCRIPTION

The new variety 'CONSTITUTION' has not been observed under all possible environmental conditions. The

characteristics of the new variety 'CONSTITUTION' may vary in detail, depending upon variations in environmental factors, including weather (temperature, humidity and light intensity), day length, soil type and location. In addition, the characteristics of any parental variety or comparison variety included in Table 1 of the present invention may vary in detail, depending upon variations in environmental factors, including weather (temperature, humidity and light intensity), day length, soil type and location.

The aforementioned photographs, together with the following description of the new variety 'CONSTITUTION', unless otherwise noted, are based on observations taken during the 2020-2021 growing season in Hillsborough County, Fla. These measurements and ratings were taken from plants of 'CONSTITUTION' dug from a high-elevation nursery located in Siskiyou County, Calif. during mid-September 2020 and planted approximately four to five days later in Hillsborough County, Fla. The approximate age of the observed plants is four months. Yield observations including average weight and marketable yield, along with fruit quality characteristics including soluble solids, were measured during the 2020-2021 growing season. Flower measurements and characteristics are from secondary flowers unless otherwise noted. Fruit characteristics and measurements are from secondary fruit, unless otherwise noted.

Where noted, color terminology follows The Royal Horticultural Society Colour Chart, London (2015).

The following characteristics describe fruit, plant, stolon, foliage, fruiting truss, flower, reproductive organs and pest and disease characteristics of the new strawberry 'CONSTITUTION'.

Fruit characteristics:

Color of mature fruit.—RHS 46B.

Color of internal flesh (excluding core).—RHS 42A.

Color of core.—RHS 49D.

Average length (cm).—4.2.

Average width (cm).—3.3.

Size.—Large.

Average length/width ratio.—1.3 (ranges from as long as broad to slightly longer than broad).

Average calyx diameter (cm).—4.3.

Season average weight (gm).—21.1.

Achene color, shaded side.—RHS 183C.

Achene color, sun-exposed side.—RHS 153B.

Average achene weight (mg).—0.6.

Average achenes per berry.—457.

Average achene length (mm).—2.2.

Average achene width (mm).—0.9.

Season marketable yield (gm/plant).—273.

Predominant shape.—Conical.

Difference in shape between primary and secondary fruit.—Ranges from moderate to large.

Band without achenes.—Narrow.

Evenness of surface.—Even.

Evenness of color.—Even or very slightly uneven.

Glossiness.—Ranges from medium to strong.

Insertion of achenes.—Even to slightly below surface.

Position of calyx attachment.—Inserted.

Attitude of sepals.—Outward.

Size of calyx in relation to fruit diameter.—Larger.

Adherence of calyx (when fully ripe).—Strong.

Firmness of flesh.—Firm.

Distribution of red color of the flesh.—Marginal and mildly central.

Hollow center expression.—Minimal.

Average cavity length (mm).—9.3.
Average cavity width (mm).—2.5.
Time of first flowering.—Early (early to mid-October in Hillsborough County, Fla.).
Flowering season.—October-February. 5
Time of first fruit.—Early (mid-November in Hillsborough County, Fla.).
Fruiting season.—November-March.
Post-harvest fruit longevity.—9-11 days if stored according to industry standards. 10
Type of bearing.—Not remontant.
 Plant characteristics:
Average height (cm).—20.2.
Average spread (cm).—34.6. 15
Size.—Large.
Habit.—Upright.
Density.—High.
Vigor.—High.
 Stolon characteristics: 20
Color.—RHS 145A (yellow green group).
Anthocyanin coloration.—RHS 176B (greyed orange group).
Anthocyanin intensity.—Moderately intense.
Pubescence.—Moderately Dense. 25
Attitude of hairs.—Slightly outward.
Average quantity in nursery (per square foot).—13.5.
Average diameter at the bract (mm).—2.2.
 Terminal leaflet characteristics:
Color of upper surface.—RHS 147A (medium yellow green). 30
Color of underside.—RHS 147B (yellow green group).
Average length (cm).—7.3.
Average width (cm).—6.75.
Average area terminal (cm²).—49.3.
Average length/width ratio.—1.08 (longer than broad). 35
Shape of base.—Obtuse.
Margins (shape of teeth).—Serrate to crenate.
Average serrations per leaf.—20.1.
 Foliage characteristics: 40
Color of upper surface.—RHS 147A (medium yellow green).
Color of underside.—RHS 147B (yellow green group).
Number of leaflets.—3.
Leaf size.—Small to medium. 45
Average length (cm).—9.15.
Average width (cm).—11.80.
Average area foliage (cm²).—107.97.
Shape in cross section.—Slightly concave.
Texture/interveinal blistering.—Ranges from light to medium. 50
Leaf glossiness.—Medium.
Leaf variegation.—Absent.
Venation pattern.—Pinnate/cross venulate.
Apex descriptor.—Obtuse. 55
 Petiole characteristics:
Petiole color.—RHS 144D (yellow green group).
Average length (cm).—11.65.
Average diameter (mm).—1.88.
Petiolule color.—RHS 144D (yellow green group). 60
Petiolule average length (mm).—5.43.
Average petiolule diameter (mm).—1.3.
Attitude of hairs.—Slightly outward.
Texture.—Moderate to smooth.
Frequency of bract leaflets.—Ranges from 0 to 2 (80% occurrence). 65

Size of bract leaflets (mm²).—5.3.
Pubescence.—Moderate.
 Stipule characteristics:
Color.—RHS 145C (yellow green group).
Anthocyanin coloration.—RHS 182C (greyed red group).
Anthocyanin intensity.—Moderate.
Average length (mm).—30.1.
Average width (mm).—8.34.
Base descriptor.—Truncate.
Apex descriptor.—Obtuse.
Shape.—Triangular.
Margin.—Smooth.
Texture.—Moderate to smooth. 15
 Fruiting truss characteristics:
Anthocyanin coloration.—147B.
Anthocyanin intensity.—Light.
Pubescence.—Light to moderate.
Attitude at first pick.—Prostrate.
Position relative to foliage.—Ranges from level with to above.
Flower quantity (average per plant season long).—20-40 (medium to high).
Average fruits per truss.—19.2.
Pedicel attitude of hairs.—Slightly outward.
Average pedicel length (cm).—21.0.
Average pedicel diameter (mm).—1.9.
Pedicel texture.—Moderate to smooth.
Pedicel color.—RHS N144B (yellow green group).
Average peduncle length (cm).—N/A.
Average peduncle diameter (mm).—N/A.
Peduncle texture.—N/A.
Peduncle color.—N/A. 35
 Flower characteristics:
Flower bud shape.—Pyriform.
Average flower bud length (mm).—18.9.
Average flower bud diameter (mm).—5.7.
Flower bud color.—RHS 145A (yellow green group).
Flower depth (mm).—17.5.
Corolla (flower) average diameter (mm).—32.1 (ranges from medium to large).
Upper petal color.—RHS NN155A (white group).
Lower petal color.—RHS NN155A (white group).
Petal shape.—Orbicular.
Petal apex descriptor.—Rounded.
Petal margin.—Smooth.
Petal base.—Decurrent.
Petal texture.—Smooth.
Petal average length (mm).—13.3.
Petal average width (mm).—11.8.
Petal average length/width ratio.—1.1 (as long as broad).
Average petals per flower.—5.9.
Relative position of petals (flowers with 5 or 6 petals).—Overlapping.
Upper sepal color.—RHS 137A (green group).
Lower sepal color.—RHS N138B (yellow green group).
Sepal shape.—Lanceolate.
Sepal apex descriptor.—Acute.
Sepal margin.—Smooth to serrate.
Sepal texture.—Moderately smooth.
Sepal average length (mm).—16.6.
Sepal average width (mm).—7.4.
Sepal average length/width ratio.—2.2.

Average sepals per flower.—11.7.

Calyx average diameter (mm).—38.7.

Size of calyx relative to corolla.—Larger.

Size of inner calyx relative to outer calyx.—Same.

Reproductive organs:

Receptacle color.—RHS 1B.

Pollen color.—RHS 15A (yellow group).

Stamen.—Present.

Average filament length (mm).—2.6.

Filament color.—RHS 145C.

Average anther length (mm).—1.5.

Anther shape.—Ovoid.

Anther color.—RHS 15A (yellow orange color).

Average pistils per flower.—457.

Pistil length (mm).—0.5-2.5.

Style length (mm).—<2.5.

Style color.—RHS 1B.

Stigma diameter (mm).—<0.5.

Stigma shape.—Simple.

Ovary color.—RHS 145B (green yellow group).

Pollen amount.—Abundant.

We claim:

- 10 **1.** A new and distinct strawberry plant named ‘CONSTI-TUTION’, as herein described and illustrated by the characteristics set forth above.

* * * * *

FIG. 1



FIG. 2



FIG. 3



FIG. 4

